

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
6 FT	(a)	(i)		Tick boxes 1,4,5: Saturn orbits the Sun with a velocity of 5 km/s faster than Pluto (1) Mars and the Earth have the same day length (1) The Earth has the greatest density (1) 4 boxes (2) 5 boxes (1) all boxes (0)			3	3		
FT	(b)	(i)		Longer wavelength						
		(ii)		Frequency increases						
		(iii)		Moves <u>away faster</u> Do not accept 'further away'						
1 HT	(c) FT (a) HT	(i)		Temperature = within range of -109 to -66(1) orbital time = within range 688 to 4330 (1)			2	2	2	
		(ii)		{Mass / diameter} smaller than the moon / smallest mass / smallest diameter accept 'smallest size' do not accept 'it is small/smaller'			1	1	1	
		(iii)		Scales: x-axis in intervals of 1/ 2 cm and y-axis in intervals of 10 / 2 cm (1) 5 plots correct $\pm$ less than one square tolerance (2) 4 plots correct $\pm$ less than one square tolerance (1) 3 or fewer plots correct $\pm$ less than one square tolerance (0) smooth curve (1) ignore extrapolation Do not accept thick/sketchy/wispy lines Do not award marks if incorrect variables are plotted or planets in incorrect order or bar chart If x-axis is labelled 'planets' do not award scale mark Moon plotted = neutral		4		4	4	
		(iv)		Orbital velocity decreases with distance / this graph curves downwards(1) (To be proportional) line must be a straight, upward and through origin (1)		2		2	2	